

D498

Data Analysis with R

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A new analyst at a bike-share program with 300 stations receives a year of trip data. Her manager, who used to work in finance, wants a regression model by Friday. What should she do first?

- A. Build the model the way finance teams do.
- B. Explore the data: summaries, distributions, missing values, and plots.
- C. Fit the regression model immediately.
- D. Run a formal hypothesis test.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

In R, a script for a school district with 8 high schools defines `x <- c(10, 20, 30, 40)`. What does `x[2]` return?

- A. 30
- B. 10 30 40
- C. 20
- D. An error, because R indexing always starts at 0.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Given `x <- c(5, 9, 2, 7)`, what does `x[-2]` return?

- A. 9
- B. 5 2 7
- C. 2
- D. An error, because negative indexes are not allowed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A column of daily temperatures from 60 weather stations has 3 values stored as NA. The analyst runs `mean(temp)`. The stations were installed in 2015. What is returned?

- A. The mean of the non-missing values.
- B. An error.
- C. NA
- D. 0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

`x <- c(4, NA, 8)`. Which call returns 6?

- A. `mean(x, na.rm = TRUE)`
- B. `sum(x, na.rm = TRUE)`
- C. `mean(is.na(x))`
- D. `mean(x)`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Using `dplyr` on a sales data frame with 45 columns from a furniture maker, an analyst wants only the rows where region is West. The data frame was built by a contractor. Which call does this?

- A. `mutate(df, region = 'West')`
- B. `select(df, region)`
- C. `filter(df, region == 'West')`
- D. `arrange(df, region)`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

An analyst needs a data frame with just the price and qty columns, keeping every row. Which call does this?

- A. `df[c('price', 'qty'), ]`
- B. `summarise(df, price, qty)`
- C. `filter(df, price, qty)`
- D. `select(df, price, qty)`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An analyst at a coffee chain with 140 cafés wants to compare the spread and median of daily sales across five regions, including outliers. The chain's logo was redesigned this year. Which plot fits best?

- A. One histogram of all daily sales combined.
- B. A scatter plot of sales against date.
- C. A pie chart of total sales by region.
- D. Side-by-side boxplots, one per region.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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9.

summary() on delivery times gives Min 5, 1st Qu. 18, Median 25, Mean 31, 3rd Qu. 38, Max 120. The data came from 2,000 deliveries in one city. What is the IQR?

- A. 20
- B. 13
- C. 115
- D. 6

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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10.

An analyst for a regional airline with 70 aircraft writes `ggplot(df, aes(x = distance, y = fare)) + geom_point(aes(color = 'blue'))`. The points come out salmon-colored with a legend that reads “blue.” What fixes it?

- A. Spell it colour inside aes().
- B. Set the color outside aes(): `geom_point(color = 'blue')`.
- C. Move `aes(color = 'blue')` into the `ggplot()` call.
- D. Remove `aes()` from the `ggplot()` call.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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11. [TRANSFER — reversed stem / different variable]

Which layer colors points by the airline variable, giving one color per airline and a legend?

- A. `geom_point(aes(color = airline))`
- B. `geom_point(color = airline)`
- C. `geom_point(aes(fill = airline))`
- D. `geom_point(aes(color = 'airline'))`

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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12.

A histogram of household incomes from a survey of 5,000 homes is piled up on the left with a long tail to the right. The survey was run by phone. Which EDA step best reveals the shape of the bulk of the data?

- A.** Square the incomes.
- B.** Delete every income above the mean.
- C.** Re-run the survey online.
- D.** Plot income on a log scale.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A new analyst at a bike-share program ~~with 300 stations~~ receives a year of trip data. Her manager, ~~who used to work in finance~~, wants a regression model by Friday. What should she do first?

Target word: first step **Decides:** The data has not been explored yet.

A	Camouflage	The manager's background does not set the process.
B	Correct	EDA reveals structure and problems before modeling.
C	Dropped element	Skips learning whether the data is fit to model.
D	Adjacent category	Confirmatory testing comes after exploration.

2. Answer C Concept 2

In R, a script ~~for a school district with 8 high schools~~ defines `x <- c(10, 20, 30, 40)`. What does `x[2]` return?

Target word: value at index 2 **Decides:** R indexes from 1.

A	Confusion set	Counts from 0, as Python does.
B	Inverted operation	That is <code>x[-2]</code> , which removes element 2.
C	Correct	The second element.
D	Absolute language	R starts at 1.

3. Answer B Concept 2 · transfer

Given `x <- c(5, 9, 2, 7)`, what does `x[-2]` return?

Target word: negative index **Decides:** A negative index removes that position.

A	Inverted operation	Keeps only the element that should be removed.
B	Correct	Removes the second element, 9.
C	Confusion set	Python-style: second from the end.
D	Absolute language	R allows them for exclusion.

4. Answer C Concept 3

A column of daily temperatures ~~from 60 weather stations~~ has 3 values stored as NA. The analyst runs `mean(temp)`. ~~The stations were installed in 2015.~~ What is returned?

Target word: result of mean with NA **Decides:** `na.rm` is not set.

A	Dropped element	Needs <code>na.rm = TRUE</code> .
B	Confusion set	R returns NA; it does not stop.
C	Correct	Any NA makes <code>mean()</code> return NA by default.
D	Inverted operation	NA is not treated as zero or as an empty result.

5. Answer A Concept 3 · transfer

`x <- c(4, NA, 8)`. Which call returns 6?

Target word: choose the call — reverse **Decides:** Mean of 4 and 8 with NA removed.

A	Correct	<code>(4 + 8) / 2 = 6</code> .
B	Adjacent category	Returns 12.
C	Confusion set	Returns the share missing, about 0.33.
D	Dropped element	Returns NA.

6. Answer C Concept 4

Using `dplyr` on a sales data frame with 45 columns from a furniture maker, an analyst wants only the rows where region is West. ~~The data frame was built by a contractor.~~ Which call does this?

Target word: row subset **Decides:** Keep rows meeting a condition.

A	Inverted operation	Overwrites every row's region instead of filtering.
B	Confusion set	<code>select()</code> keeps columns, not rows.
C	Correct	<code>filter()</code> keeps matching rows.
D	Adjacent category	Sorts rows but keeps all of them.

7. Answer D Concept 4 · transfer

An analyst needs a data frame with just the price and qty columns, keeping every row. Which call does this?

Target word: column subset — reverse **Decides:** Keep columns, all rows.

A	Inverted operation	Puts the names in the row position.
B	Adjacent category	Collapses rows into summaries.
C	Confusion set	<code>filter()</code> subsets rows by conditions.
D	Correct	<code>select()</code> keeps named columns.

8. Answer D Concept 5

An analyst at a coffee chain with 140 cafés wants to compare the spread and median of daily sales across five regions, including outliers. ~~The chain's logo was redesigned this year.~~ Which plot fits best?

Target word: plot type **Decides:** Compare distributions across groups.

A	Dropped element	Drops the region grouping.
B	Adjacent category	Shows a time pattern, not a group comparison.
C	Confusion set	Shows shares, not spread or median.
D	Correct	Shows median, spread, and outliers for each group.

9. Answer A Concept 6

summary() on delivery times gives Min 5, 1st Qu. 18, Median 25, Mean 31, 3rd Qu. 38, Max 120. ~~The data came from 2,000 deliveries in one city.~~ What is the IQR?

Target word: IQR **Decides:** Q3 – Q1.

A	Correct	$38 - 18 = 20$.
B	Dropped element	Q3 – median drops Q1.
C	Adjacent category	Max – min is the range.
D	Wrong stem number	Mean – median, a skew hint, not the IQR.

10. Answer B Concept 7

An analyst for a regional airline ~~with 70 aircraft~~ writes `ggplot(df, aes(x = distance, y = fare)) + geom_point(aes(color = 'blue'))`. The points come out salmon-colored with a legend that reads “blue.” What fixes it?

Target word: fix for a constant color **Decides:** A constant was mapped inside aes().

A	Confusion set	Both spellings work; it is still a mapping.
B	Correct	Outside aes() sets a constant; no legend is created.
C	Adjacent category	Still maps a constant string.
D	Dropped element	Drops the x and y mappings.

11. Answer A Concept 7 · transfer

Which layer colors points by the airline variable, giving one color per airline and a legend?

Target word: mapping a variable — reverse **Decides:** Variables are mapped inside aes().

A	Correct	Maps the variable to color.
B	Inverted operation	Setting outside aes() expects a constant, not a column.
C	Adjacent category	Default points use color, not fill.
D	Confusion set	Maps the literal text, giving one color.

12. Answer D Concept 8

A histogram of household incomes ~~from a survey of 5,000 homes~~ is piled up on the left with a long tail to the right. ~~The survey was run by phone.~~ Which EDA step best reveals the shape of the bulk of the data?

Target word: transformation **Decides:** Strong right skew.

A	Inverted operation	Stretches the tail further.
B	Dropped element	Throws away real data.
C	Camouflage	Collection method does not change income skew.
D	Correct	Log spreads the bulk and compresses the tail.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Purpose of EDA	Exploratory data analysis summarizes and plots the data to learn its structure, problems, and patterns before any modeling or formal testing.	Jumping straight to a model or a hypothesis test.	A new data set and “what should she do first?”
2	R indexing	R vectors start at index 1; a negative index removes that element.	Using Python habits: counting from 0 or treating negatives as counting from the end.	x[2] or x[-2] on a short vector.
3	NA handling	Most summary functions return NA if any value is NA unless na.rm = TRUE is set.	Expecting mean() to skip NAs automatically.	A vector with NA passed to mean() or sum().
4	filter vs. select	dplyr filter() keeps rows that meet a condition; select() keeps columns.	Swapping the two.	“Only the rows where ...” vs. “only these columns.”
5	Choosing an EDA plot	Histogram for one numeric distribution; side-by-side boxplots to compare a numeric variable across groups; scatter plot for two numeric variables.	Using a pie chart or a single histogram to compare groups.	“Compare spread and median across groups.”
6	Reading summary()	summary() reports min, Q1, median, mean, Q3, and max; IQR = Q3 – Q1, and a mean above the median suggests right skew.	Using max – min as the IQR.	A summary() printout and a spread question.
7	Mapping vs. setting in ggplot2	Inside aes() you map a variable to an aesthetic; outside aes() you set a constant.	Putting a constant like 'blue' inside aes().	A color that shows up wrong with an odd legend.
8	Transforming skewed data	A log scale spreads out the bulk of right-skewed data and compresses the long tail.	Deleting large values, or applying a transformation that stretches the tail further.	A histogram piled on the left with a long right tail.

Course scope (public description): Data Analysis with R focuses on exploratory data analysis (EDA) utilizing R. EDA is an approach for summarizing and visualizing the important characteristics of a data set. In this course you will develop skills in R programming to acquire and load data sets, create appropriate statistical summaries of data, and create data visualizations to help uncover and communicate insights about data using R.